

KARUIZAWA, Japan

WMO: 476220

Lat: 36.3419N

Lon: 138.5472E

Elev: 3283

StdP: 13.03

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	14.0	3.9	7.6	24.2	6.7	8.7	24.2	13.6	31.2	12.1	31.2	3.6	0	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	83.5	71.3	81.1	70.6	78.5	69.4	73.1	79.8	71.9	78.0	70.8	76.3	5.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	130.5	76.3	70.1	125.4	75.0	69.1	121.0	73.7	39.3	80.1	38.1	78.3	37.0	76.6	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.0	9.1	DB	5.0	87.4	3.0	1.7	2.9	88.6	1.1	89.6	-0.6	90.6	-2.8	91.9	(4)
(5)			WB	4.5	75.7	3.0	0.9	2.3	76.3	0.5	76.8	-1.2	77.3	-3.4	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	26.3	27.7	34.6	45.0	54.8	61.5	69.4	70.8	63.0	52.1	41.5	31.8
	DBStd	16.33	5.12	6.09	6.70	7.00	5.29	4.97	4.62	4.14	5.85	5.60	5.73	5.60
	HDD50	2902	736	623	477	178	18	0	0	0	0	41	263	566
	HDD65	6498	1201	1043	941	599	319	127	16	9	105	401	706	1031
	CDD50	2289	0	0	1	29	165	345	602	644	391	105	7	0
	CDD65	410	0	0	0	0	1	22	153	188	46	0	0	0
	CDH74	1825	0	0	0	5	39	93	716	838	133	1	0	0
	CDH80	310	0	0	0	0	2	7	136	155	10	0	0	0
Wind	WSAvg	4.0	4.8	4.7	4.8	4.7	4.2	3.5	3.3	3.2	3.3	3.5	4.0	4.6
Precipitation	PrecAvg	47.5	1.5	1.6	2.5	3.1	4.0	5.7	6.8	5.9	7.7	5.0	2.4	1.3
	PrecMax	58.7	3.7	3.3	4.7	6.9	8.7	9.3	14.6	11.9	16.4	14.7	6.7	3.7
	PrecMin	30.2	0.3	0.6	0.9	1.0	1.5	2.7	3.0	2.2	1.4	0.2	0.2	0.2
	PrecStd	7.3	1.0	0.9	1.0	1.5	1.7	1.8	2.9	2.7	3.4	3.2	1.5	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	54.4	63.8	73.1	78.5	80.8	86.4	86.7	81.5	72.4	63.7	53.7
		MCWB	40.3	43.6	46.4	54.2	59.3	66.8	71.0	71.2	70.4	62.2	52.0	45.2
	2%	DB	43.0	47.5	56.4	66.9	73.7	76.5	83.7	83.8	77.8	67.7	58.9	48.8
		MCWB	35.9	39.1	43.3	50.8	56.9	64.8	71.4	71.7	68.8	60.1	49.9	41.8
	5%	DB	39.1	42.3	51.8	62.9	70.1	73.8	81.1	81.5	74.7	64.3	55.0	45.4
		MCWB	32.7	35.1	41.4	48.9	55.4	63.6	70.8	71.3	67.1	57.5	47.9	39.0
	10%	DB	35.6	38.4	47.4	58.8	66.9	70.9	78.3	79.0	71.6	61.3	51.7	41.8
		MCWB	30.2	32.5	39.4	47.8	54.5	62.4	69.9	70.7	66.1	56.1	46.2	36.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	46.1	49.5	58.9	64.3	70.2	74.9	75.0	71.8	65.0	55.6	48.5
		MCDB	45.8	52.4	57.4	63.6	70.0	76.5	82.2	82.0	78.1	69.2	59.9	50.3
	2%	WB	36.7	40.2	46.2	54.8	61.1	67.9	73.1	73.5	70.0	62.1	52.2	43.3
		MCDB	41.4	45.6	52.3	62.4	68.3	73.1	79.9	80.1	75.1	65.8	55.7	47.0
	5%	WB	33.3	36.2	43.1	51.8	58.8	66.0	71.8	72.4	68.7	59.6	49.8	39.9
		MCDB	38.1	40.6	49.9	58.3	65.4	70.7	78.0	78.5	72.8	62.4	52.7	43.7
	10%	WB	30.8	33.1	40.1	49.3	56.8	64.3	70.6	71.3	67.3	57.3	47.5	36.9
		MCDB	34.9	37.6	46.2	55.6	63.2	68.9	76.1	76.9	70.6	59.8	50.9	41.1
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.9	20.5	22.4	21.1	15.8	14.4	15.0	14.0	16.3	18.4	17.3
		MCDBR	21.5	24.1	27.4	28.8	27.9	21.1	19.4	19.3	17.2	19.7	22.8	21.6
		MCWBR	15.0	16.1	16.1	14.9	13.6	10.5	9.2	9.0	9.0	11.7	14.2	14.6
	5% WB	MCDBR	19.1	21.5	23.9	22.4	22.5	16.3	17.3	16.9	15.0	15.8	18.2	19.2
		MCWBR	14.0	15.4	16.0	13.9	12.7	9.0	8.9	8.6	8.3	10.3	12.8	14.3
Clear-Sky Solar Irradiance	taub	0.314	0.348	0.402	0.444	0.469	0.504	0.500	0.504	0.438	0.390	0.341	0.315	
	taud	2.238	2.139	2.078	2.056	2.038	2.007	2.095	2.074	2.228	2.292	2.385	2.352	
	Ebn at Noon	275	278	272	266	261	251	251	247	259	261	265	268	
	Edh at Noon	33	41	48	52	54	56	51	51	41	35	29	28	
All-Sky Solar Radiation	RadAvg	755	996	1278	1523	1676	1513	1491	1492	1205	1000	795	677	
	RadStd	38	64	81	124	166	102	209	141	104	98	60	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page